VAISHNAVI BHARADWAJ

Advanced Application Engineering Analyst

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EDUCATION

Year	Degree	CGPA	Institute
2022	B.Tech (Electronics and Instrumentation)	9.14	Haldia Institute Of Technology, West Bengal
2026	MBA (Business Analytics)	9.31	Manipal Academy of Higher Education, Karnataka

WORK EXPERIENCE

Accenture – Advanced Application Engineering Analyst

Project: AT&T – Telecommunications, US (Infra, API Monitoring, Production Support) **[Jul 2023 – Dec 2025]**

- Enhanced Linux/Unix server reliability via health checks, log diagnostics, and automated service recovery, improving uptime by approx. **15%+**.
- Engineered **C2W** workflows & API observability using AppDynamics, Dynatrace, Grafana, improving alert triaging and reducing MTTR by **25%**.
- Resolved **200+** kiosk tickets, achieving **30%** faster turnaround; delivered AT&T flagship (iPhone & Samsung) sales by **Power BI** KPI dashboards.

Project: ALLIANZ - Insurance, Spain (Backend Development) **[Jan 2026 – Present]**

- Developed insurance app using **D3 Pick**, implementing business logic, data handling, and transaction workflows on a MultiValue database.
- Architected **Multiasistencia** integrations to optimize screening pipelines, leveraging indexed structures & query tuning to improve throughput.

PROJECTS

Data Science Learning Hub | **Full Stack** | **Demo** **[Aug 2025 - Present]**

- Built **Ed-Tech SPA** across **12** subjects and **70+** modules; **15+** visualisers covering Minimax, Alpha-Beta, K-Means, SVM, PCA, BFS-DFS, etc.
- Built **adaptive quiz engine** with spaced-repetition mastery scoring, exponential time-decay weighting, and offline IndexedDB progress tracking.
- Delivered full GATE DA syllabus with KaTeX LaTeX rendering, live Mermaid diagram generation & authentic timed auto-submit exam simulation.
- Architected **offline-first code-split** SPA with all tool bundles lazy-loaded on demand, delivering instant startup without any server dependency.

Research Project | **Indian School of Business, Hyderabad** **[Mar 2022- May 2022]**

- Diameter at Breast Height (DBH) estimation of trees in videos using advanced deep learning–based analytical techniques

- Utilized Laplacian variance for frame extraction achieving **95%+** clear frames and initiated tree trunk masking with Segment Anything Model.
- Enhanced masking by integrating **SAM** with confidence thresholding, area analysis, & developed PCA-based algorithm for outline delineation.
- Evaluated depth estimation models **(EVP)**, explored stereo vision techniques with IMU sensors to improve distance measurement accuracy.

Machine Learning in satellite Imagery to monitor health in low and middle-income countries **[May 2023 - September 2023]**

- Trained boosting models on satellite imagery data, conducting comprehensive exploratory data analysis **(EDA)**, achieving an **RMSE** of **11.39**.
- Employed XGBoost to identify the top **200** & **500** important features, optimizing model performance and reducing RMSE to 11.257 & 11.292.
- Improved loss by imputing data for missing points using mean & median, further refining the **LightGBM** model to achieve final RMSE 11.6.

INTERNSHIP

Java Full Stack + GCP Training | **Wipro** **[Apr 2022 – Sep 2022]**

- Completed full-stack training covering backend services, frontend frameworks, cloud workflows, and modern software development practices.
- Integrated Teachmint Admin Portal with EventBeep using GCP Functions, Cloud Storage, enabling smooth data handling across both platforms.
- Built MEAN-stack e-commerce application with product listings, cart logic, admin features, demonstrating complete CRUD workflow capability.

Automation Projects | **Cognizant** **[May 2022 - Aug 2022]**

- Built UI automation suites using Selenium + Java for the Live Support GSD Portal, Display Bookshelves app, and Online Book Search platform.
- Improved test coverage by **11%** & reduced manual regression effort through automated workflows, enabling faster and reliable release cycles.

ACHIEVEMENTS

- Served as **Joint Secretary** for the ISA and ISOI student chapters in 2021, contributing actively to events and organizational initiatives.
- Secured **1st** place in the **CIRCUITRIX** hardware competition organized by the **IEEE** Society in 2019, demonstrating strong technical skills.
- Completed a **Senior Diploma** in **Kathak** from Allahabad University in 2017, showcasing dedication to classical dance training.

COURSEWORK INFORMATION

- **Computer Science:** Programming & Data Structures | Algorithms Laboratory
- **Electronics & Instrumentation Engineering:** Microprocessor 8085 and 8086 | VLSI | Digital Computing | Analog | Optical instrumentation
- **Mathematics:** Probability & Statistics, Linear Algebra, Numerical & Complex Analysis, Transform Calculus, Advanced Calculus
- **Artificial Intelligence:** Statistical Learning | Machine Learning Foundations & Applications | Deep Learning Foundations & Applications
- **MOOC:** Google Cloud Big Data & Machine Learning Fundamentals | Blockchain, Solidity & Smart Contracts | Data Security & Code of Ethics

SKILLS AND EXPERTISE

- **Programming Languages:** Java, JavaScript, TypeScript, Python, SQL, C, C++, R, C#, Kotlin, Swift, MATLAB, Bash, HTML
- **Frameworks & Libraries:** React, Vite, Tailwind CSS, KaTeX, Angular, Spring Boot, Express, Node.js, Selenium WebDriver
- **Cloud & DevOps:** Google Cloud Platform, Azure, Linux/Unix Administration, AppDynamics, Dynatrace, Grafana, Microservices, REST APIs
- **Developer Tools:** Power BI, VS Code, Git, GitHub, Postman, ServiceNow, JIRA, Automation Scripting (Shell/Python), MySQL Workbench
- **Core Competencies:** Production Support, API Monitoring, Server Operations, Troubleshooting & RCA, UI Automation, Documentation

CERTIFICATIONS

Salesforce Certified Administrator | Microsoft Azure Fundamentals | GCP ACE certification | GCP Digital Cloud Leader | MongoDB SI Associate Certification | Introduction to Internet Of Things by NPTEL | Oracle Fusion AI Agent Studio Foundations Associate | Oracle AI Foundations Associate | Machine Learning and Deep Learning Specialization by Andrew Ng, Coursera | Introduction to Algorithms, MIT OpenCourseWare